

NCME MISSION FUND PROPOSAL

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Descriptive Title (Maximum: 15 words)

Collaborative Resources and Training for AI-Assisted Q-Matrix Construction in Classroom Assessment

BODY (MAXIMUM: 1500 WORDS)

Purpose

The purpose of this project is to bridge the gap between psychometric research and classroom practice by developing collaborative, practitioner-oriented resources for AI-assisted Q-matrix construction in cognitive diagnostic (CD). Although recent methodological advances, such as the development of the nonparametric CD (NPCD) framework (e.g., Chang et al., 2019; Chiu & Chang, 2021; Chiu & Douglas, 2013; Chiu & Köhn, 2019; Chiu et al., 2018; Wang et al., 2023), have increased the technical feasibility of CD assessment in small instructional settings, adoption in classroom practice remains limited. The primary issue is the lack of accessible resources for Q-matrix construction in practice. Traditional approaches, such as expert workshops, one-on-one consultations, are resource-intensive and difficult to scale, while AI-assisted Q-matrix construction algorithms offer a promising pathway to overcome these limitations. However, realizing this potential in practice requires more than the availability of AI algorithms alone; educators require clear guidance and shared standards to ensure AI is used responsibly and transparently.

Aligned with NCME President Amy Hendrickson's themes of service and collaboration, this project seeks to translate emerging research in AI-assisted Q-matrix construction into practitioner-informed resources and tools to support educators, assessment practitioners, and researchers in implementing CD assessments. Project deliverables include interactive web-based applications (e.g., shiny R apps), annotated examples, and hands-on training materials.

Theoretical Framework

The project is grounded in the NPCD framework, which bypasses parametric modeling and does not require large calibration samples, making it well suited for classroom assessments with small numbers of students. In all CD analyses, the Q-matrix (Tatsuoka, 1985) serves as the foundational component that indicates the associations between test items and the latent attributes they are intended to measure. However, Q-matrix construction relies heavily on expert judgment, making it labor-intensive, subjective, and difficult to scale across items or assessments. A further complication arises from the fact that data-driven methods for Q-matrix construction (e.g., Chen et al., 2015; He et al., 2023; Köhn et al., 2025; Xu & Shang, 2018) are not applicable in small assessment settings. To address the challenges, this project adopts an AI-assisted workflow for Q-matrix construction, termed the Traceable Agent-Committee Encoding for Q-matrix construction (TRACE-Q; Xiao & Chiu, in development). TRACE-Q is a recent methodological development that builds on a structured multi-agent large language model (LLM) framework, aiming to enhance accessibility, scalability, stability, and computational tractability. The workflow is data-independent, which makes it particularly suitable for small-scale instructional settings, such as classrooms. Specifically, TRACE-Q employs multiple independent AI agents that each analyze items and propose q-vectors accompanied by explicit rationales citing specific item features. Because agents work independently, practitioners can examine where mappings converge (suggesting robustness) or diverge (flagging items needing closer review). In the subsequent expert consolidation phase, these AI-generated drafts are reviewed and refined by human experts to ensure validity. A pilot implementation of TRACE-Q using Tatsuoka's (1990) fraction-subtraction assessment demonstrated improved model fit relative to two established benchmarks. Although these initial findings are encouraging, the current project is necessary to systematically evaluate the pipeline's stability and identify potential failure modes across a broader range of STEM disciplines. Such an evaluation will transform a successful pilot into a robust, practitioner-ready resource for classroom assessment.

Methodology

Methodological activities are organized into three interconnected components:

1. The research team will develop interactive open-access shiny R web applications that operationalize TRACE-Q in an end-to-end workflow for mapping assessment items to cognitive attributes. The applications will allow users to explore the proposed q-vector for each item along with supporting rationales, enable side-by-side comparison of alternative mappings across agents or runs, and provide revision and documentation features (e.g., edit history and reviewers' notes) to facilitate expert consolidation.
2. The project will produce a suite of instructional training materials co-developed with practitioners. Parallel to the shiny apps, these materials include step-by-step user guides, annotated case studies based on classroom assessments, and practical decision aids to help educators construct Q-matrices under common instructional constraints (e.g., small samples and limited or no response data). Draft materials will be iteratively refined in collaboration with educators and assessment specialists to ensure accuracy, relevance and usability across diverse instructional contexts.
3. The project will conduct a collaborative pilot and formative evaluation involving approximately 15-20 higher-education STEM educators. Participants will engage in

hands-on activities using the developed tools and materials through structured virtual walkthroughs and guided review activities. The participants will also serve as “content experts” by reviewing the provisional Q-matrix generated by TRACE-Q. Structured feedback on usability, interpretability, transparency, and practical relevance of TRACE-Q and accompanying resources will be collected. This feedback will then be used formatively to refine the applications, instructional materials, and best-practice guidance, ensuring that project outcomes reflect practitioners’ needs and support responsible use of AI-assisted methods in classroom assessment. Recruiting approximately 15-20 STEM educators provides a feasible mechanism for collaboration and service. The participants will serve as collaborative partners by contributing practical insights and this group size enables diverse perspectives while remaining appropriate for a resource-development initiative. It is important to note that although the tools are designed for classroom assessment broadly, the pilot will focus on higher-education STEM contexts as a starting point.

Anticipated Results, Outcomes, and/or Products

1. *The TRACE-Q Interactive Web Portal*: The primary product of this work is an open-access R shiny web application that operationalizes the TRACE-Q pipeline. Unlike a “black-box” AI tool, the proposed application will produce several key outputs for the user, including (a) AI-generated Q-matrix drafts for each uploaded set of assessment items; (b) traceable rationales for each suggested q-vector, explaining why the AI agent identified specific cognitive skills for a given item; and (c) a side-by-side “committee view” that allows practitioners to see where different AI agents agree or disagree in their skill mappings.

The following figures illustrate a preliminary workflow showing how the Shiny app supports Q-matrix construction. Figure 1 introduces key concepts in cognitive diagnosis, including the role of the Q-matrix. Figure 2 illustrates the TRACE-Q workflow and highlights how traceable outputs are generated at each stage of the process. Figure 3 presents an example of the feedback survey used to collect practitioners’ input and feedback.

Figure 1

CDM 101 Tutorial Module and Q-matrix Visualization

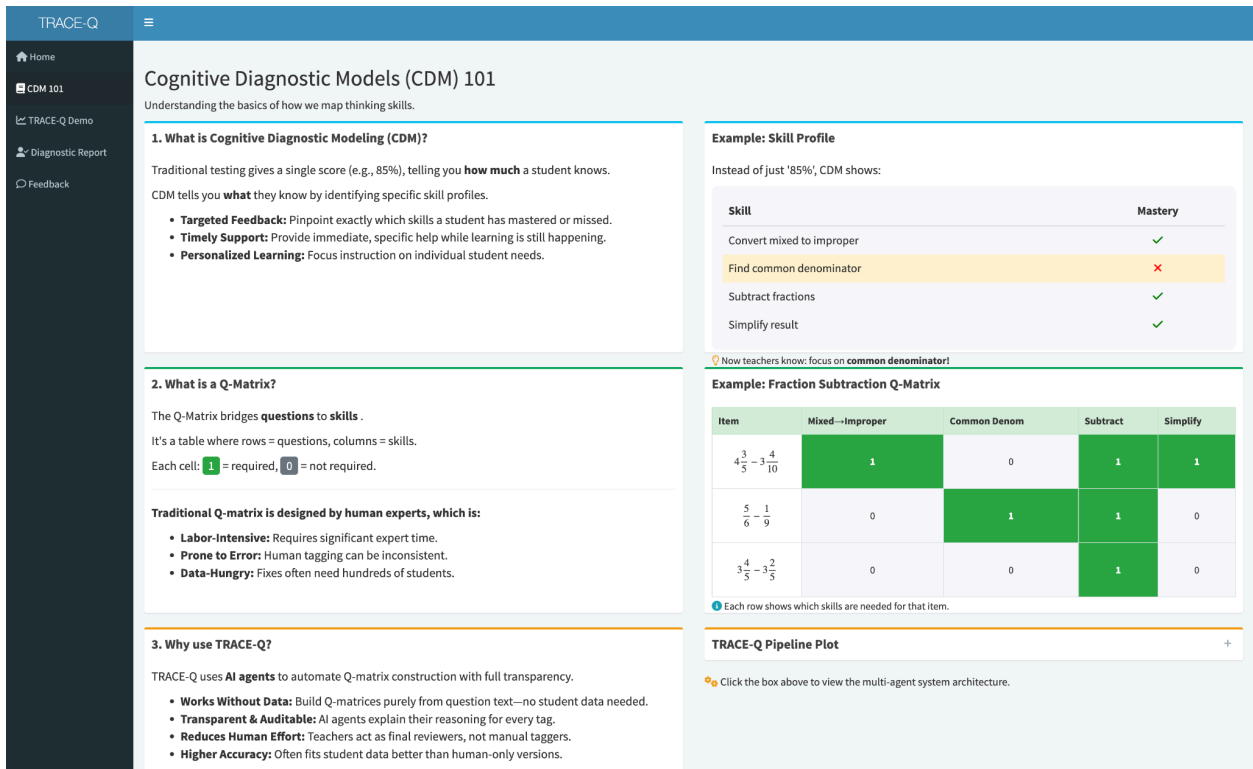


Figure 2

TRACE-Q Multi-Agent Pipeline with Traceable Rationales

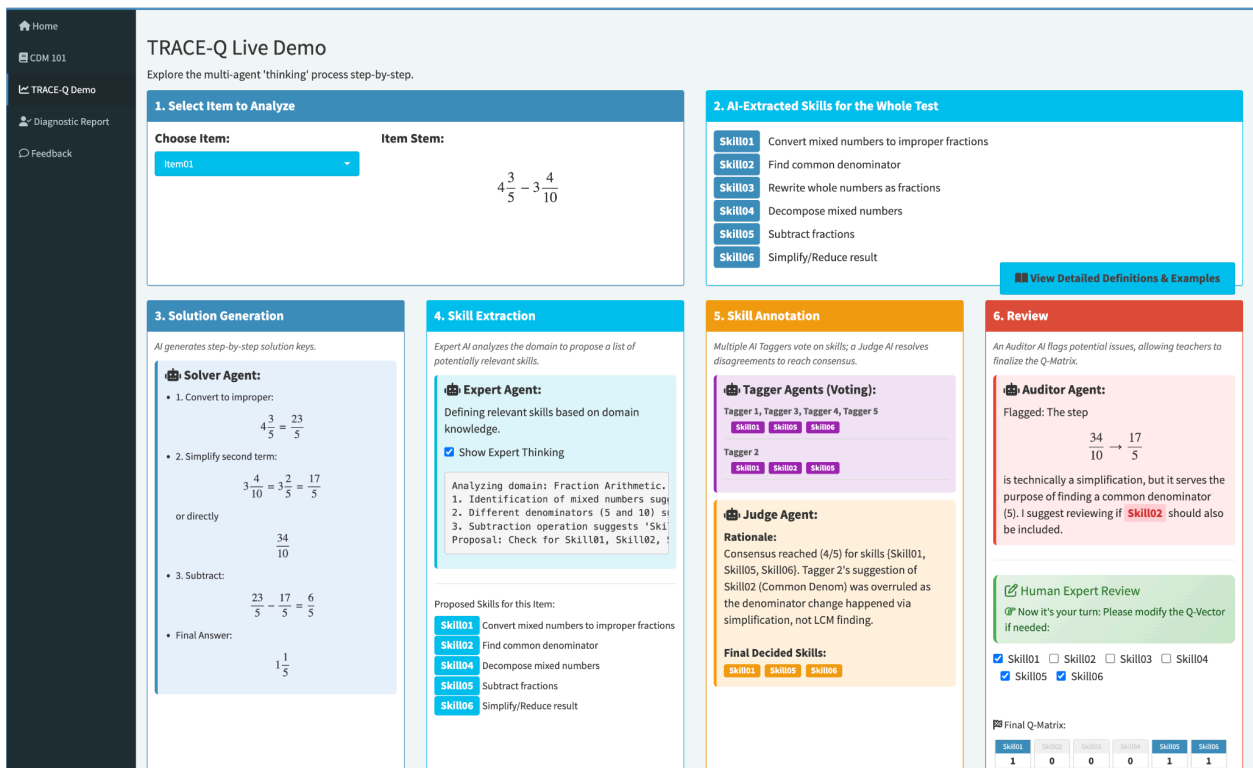


Figure 3

Practitioner Feedback Interface

2. *Practitioner-Engaged Training Toolkit*: In response to the theme of service, the project will produce a toolkit for CD assessment, including (a) best-practice guidance, providing clear protocols on when to rely on AI and when to override it with expert judgment; (b) annotated case library with a collection of 2-4 case studies that demonstrate the full end-to-end process from raw quiz items to a validated Q-matrix; and (c) structured virtual walkthroughs and guided review activities to facilitate educators' reviews.
3. *Collaborative Validation Findings*: The pilot with 15-20 higher-education STEM educators will produce formative evidence including, (a) quantitative data on the consistency of TRACE-Q outputs across different STEM disciplines; (b) experts' review data that will help identify common failure modes or misconceptions the AI may have regarding specific STEM constructs; and (c) a technical report and a proposal for the 2027 NCME or AIME Annual Meeting.

Together, these products establish TRACE-Q as a shared, collaborative infrastructure for responsible AI-assisted CD assessment.

Educational Implications of Anticipated Results

The NCME Classroom Assessment Task Force emphasized the need to develop and disseminate resources that illuminate the power and purposes of high-quality classroom assessment to support student learning and foster student ownership (Goal 2). CD was specifically developed to address this goal by providing fine-grained information about students' mastery of instructional content and by supporting timely, actionable feedback that informs learning and instruction. However, practical barriers—particularly those associated with Q-matrix construction—have limited the widespread use of CD approaches in classroom settings. This project responds to that call by repositioning CD as a scalable and usable approach to classroom assessment, rather than a method accessible only through specialized expertise.

A central anticipated result is a shift in how CD assessment is introduced and used in instructional settings. By integrating AI-assisted support with structured human review, the project transforms Q-matrix construction into a collaborative activity. This is expected to broaden participation in CD assessment while maintaining the core psychometric requirements of interpretability and validity.

The project also provides a model for the responsible incorporation of AI into classroom assessment. Rather than treating AI as an automated decision-maker, the TRACE-Q framework positions AI as a supportive and traceable partner in assessment design. An anticipated outcome is the development of clearer guidance for “Human-in-the-Loop” workflows that help educators and practitioners understand not only what decisions are made, but why they are made.

Through engagement with higher-education STEM educators, the project is expected to cultivate a shared understanding of effective AI-assisted design across research and practice communities. This collaborative process ensures that the resulting tools and guidance reflect real instructional needs and support continuous professional learning. Finally, the products of this project will serve as a sustainable infrastructure for high-quality classroom assessment and informed instructional decision-making. In doing so, the project directly advances the Mission Fund's goals of service, collaboration, and the dissemination of resources that strengthen classroom assessment practice.

References

Chang, Y.-P., Chiu, C.-Y., & Tsai, R.-C. (2019). Nonparametric CAT for CD in educational settings with small samples. *Applied Psychological Measurement*, 43(7), 543-561.

Chiu & Chang, 2021;

Chiu & Douglas, 2013;

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Tatsuoka, K. K. (1985). A probabilistic model for diagnosing misconception in the pattern classification approach. *Journal of Educational and Behavioral Statistics*, 12(1), 55-73.

Tatsuoka, K. K. (1990). Toward an integration of item-response theory and cognitive error diagnosis. In N. Frederiksen, R. Glaser, A. Lesgold, & M. G. Shafto (Eds.), *Diagnostic monitoring of skill and knowledge acquisition* (pp. 453–488). Lawrence Erlbaum Associates.

Wang et al., 2023

TIMELINE AND BUDGET

Timeline for Execution

Activity	05/2026-06/2026	07/2026-08/2026	09/2026-10/2026	11/2026-12/2026	01/2027-02/2027	03/2027	04/2027
TRACE-Q Code Consolidation	X						
TRACE-Q Evaluation	X	X					
Shiny Apps	X	X	X				
Training Materials		X	X	X			
Practice Guidance		X	X	X			
Educator Recruitment	X	X	X				
Educator activities				X	X		
Data Collections					X	X	
Data Analysis and Tool Refinement					X	X	X
Writing Report						X	X

Estimated Budget

Amount	Use of Money	Payment Schedule	Deliverables
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\$8,000	Graduate research assistance	Project Initiation (Phase 1)	Functional R shiny Beta with TRACE-Q integration.
\$4,000	Graduate research assistance	Mid-Project (Phase 2)	Data processing and workshop technical support.
\$5,000	Participant Honoraria	Phase 2 Completion	\$250 stipends for 20 STEM faculty upon completion of review and validation.
\$1,000	Cloud & API Infrastructure	Continuous	Ongoing LLM token usage and professional shiny hosting.
\$2,000	Travel & Dissemination	Project End	Technical report and 2027 NCME presentation materials.
Total: \$20,000			